SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	22-07-26	**Duration:**	60 minutes

Code of Conduct

I M.Chandana(192472250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Malli Chandana

Tourism and Hospitality Domain: Smart heritage sight information and virtual tour platform

Introduction

Tourism is one of the fastest-growing industries and plays an important role in economic growth, cultural exchange, and heritage preservation. Heritage sites such as monuments, museums, temples, forts, and archaeological locations attract millions of visitors every year. However, many of these sites still depend on traditional methods like brochures, signboards, and tour guides, which often provide limited information and fail to meet the expectations of modern tourists. Visitors also face challenges such as language barriers, outdated information, limited accessibility, and long ticket queues.

The **Smart Heritage Site Information and Virtual Tour Platform** is proposed to overcome these challenges by integrating technologies such as **Artificial Intelligence (AI)**, **Virtual Reality (VR)**, cloud computing, and mobile applications. The platform enables users to access heritage information, enjoy virtual tours, book tickets online, and receive AI-based assistance from a single application. It also helps heritage authorities manage digital content efficiently while promoting sustainable tourism and preserving cultural heritage.

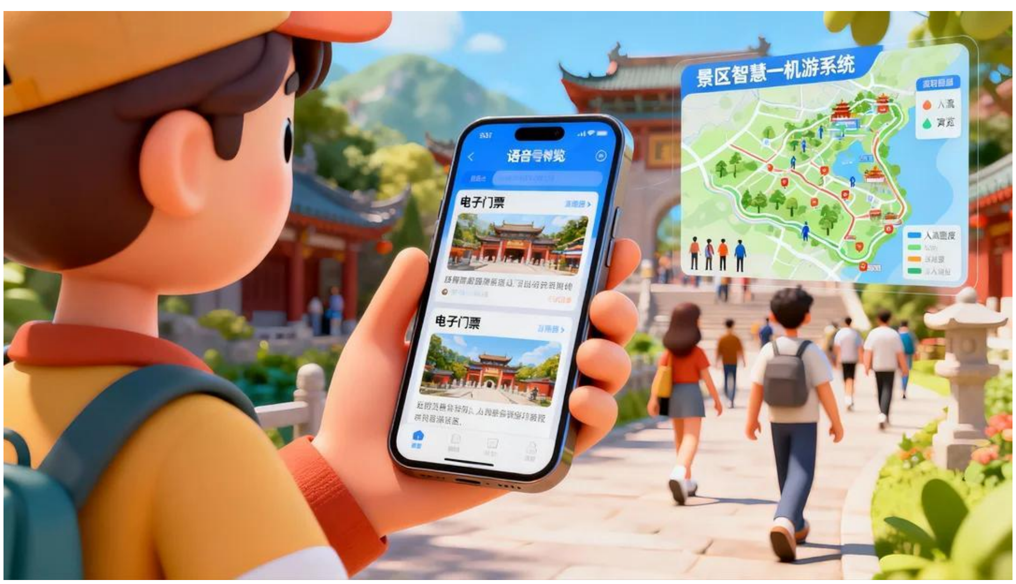

Figure 1: Concept of a Smart Heritage Site Information and Virtual Tour Platform.

1. Understand the Problem Statement

1.1 Explain the Problem Statement

Heritage sites are important cultural assets that attract tourists from around the world. However, many heritage locations still rely on traditional methods such as printed brochures, signboards, and tour guides to provide information. These methods often offer limited details, are difficult to update, and may not support multiple languages. Many tourists also face challenges such as long ticket queues, lack of personalized guidance, and limited accessibility. Existing tourism websites usually provide only basic information and do not offer interactive features like virtual tours or AI-based assistance. Therefore, there is a need for a smart digital platform that provides accurate information, virtual experiences, and convenient services while promoting cultural heritage and improving the overall tourist experience.

Key Points:

- Traditional methods provide limited and outdated information.
- Language barriers affect international tourists.
- Lack of virtual tour and AI-based assistance.
- Limited accessibility for remote and disabled users.
- Need for a centralized digital tourism platform.
- Enhances visitor experience and heritage preservation.

1.2 Objectives of the Proposed System

The main objective of the Smart Heritage Site Information and Virtual Tour Platform is to improve the tourism experience by providing a single digital platform for heritage information and services. The system offers virtual tours, online ticket booking, multimedia content, multilingual support, and AI-powered assistance. It also helps heritage authorities manage information efficiently while preserving cultural heritage digitally. The platform aims to make tourism more accessible, interactive, and convenient for users worldwide.

Key Points:

- Provide accurate heritage site information.
- Offer virtual tours and multimedia resources.
- Enable secure online ticket booking.
- Support multilingual communication.
- Improve accessibility for all users.
- Promote digital preservation of cultural heritage.

1.3 Scope of the Proposed System

The Smart Heritage Site Information and Virtual Tour Platform supports the digital management and promotion of heritage tourism. It allows users to search for heritage sites, explore historical information, take virtual tours, and book tickets online. The system also provides AI chatbot support, GPS navigation, and personalized recommendations. It is designed for tourists, heritage authorities, tourism departments, researchers, and educational institutions. The platform can be further expanded by integrating technologies such as Augmented Reality (AR), Internet of Things (IoT), and blockchain to enhance future tourism services.

Key Points:

- Covers monuments, museums, forts, temples, and heritage sites.
- Supports virtual tours and online ticket booking.
- Provides AI chatbot and GPS navigation.
- Serves tourists, authorities, researchers, and students.
- Can integrate future technologies like AR, IoT, and blockchain.
- Supports long-term digital tourism development.

2. Identify Key Stakeholders and Challenges

The success of the **Smart Heritage Site Information and Virtual Tour Platform** depends on the involvement of different stakeholders who contribute to the development, management, and use of the system. Stakeholders include tourists, heritage authorities, government departments, software developers, administrators, and local businesses. Understanding their roles and expectations helps in designing a system that meets user needs, improves collaboration, and ensures efficient heritage management. Stakeholder analysis also helps identify challenges and develop suitable software solutions.

2.1 Identify All the Stakeholders Involved

The Smart Heritage Site Information and Virtual Tour Platform involves several stakeholders who play an important role in its success. Tourists use the platform to explore heritage sites, book tickets, and access virtual tours. Heritage authorities maintain historical information and preserve cultural assets, while government tourism departments promote tourism and monitor heritage activities. Software developers and system administrators develop, manage, and maintain the platform. Educational institutions, researchers, tour guides, and local businesses also benefit from the system by using its resources and services.

Key Points:

- Tourists (Domestic and International)
- Heritage Site Management Authorities
- Government Tourism Departments
- Software Developers and System Administrators
- Tour Guides
- Educational Institutions and Researchers
- Local Businesses (Hotels, Restaurants, Transport)

2.2 Explain the Roles and Responsibilities of Each Stakeholder

Each stakeholder has specific responsibilities that support the effective functioning of the platform. Tourists use the system to search for heritage information, experience virtual tours, and book tickets. Heritage authorities update historical content and maintain accurate records. Government departments promote tourism and oversee heritage conservation. Software developers design, develop, test, and improve the platform, while administrators manage users, monitor system performance, and ensure security. Researchers, educational institutions, and local businesses use the platform to support education, tourism services, and economic development.

Key Points:

- Tourists access information, virtual tours, and booking services.
- Heritage authorities update and manage heritage content.
- Government departments promote tourism and preserve heritage.
- Developers build, test, and maintain the application.
- Administrators manage users, security, and system performance.
- Researchers and educational institutions use digital resources.
- Local businesses support tourism-related services.

2.3 Discuss the Major Challenges and Issues Faced by the Stakeholders

Despite its advantages, stakeholders face several challenges while using and managing the platform. Tourists may experience language barriers, limited internet connectivity, and difficulty accessing accurate information. Heritage authorities must regularly update content and preserve cultural authenticity. Government departments face challenges in balancing tourism growth with heritage conservation. Developers and administrators need to maintain system security, handle large amounts of data, and ensure smooth performance. Local businesses are affected by seasonal tourist demand, while researchers require reliable and updated information. Addressing these challenges through proper planning and continuous software improvements will ensure the long-term success of the platform.

Key Points:

- Language and accessibility issues for tourists.
- Regular content updates by heritage authorities.
- Balancing tourism growth and heritage conservation.
- Cybersecurity and system maintenance challenges.
- Managing large multimedia data efficiently.
- Seasonal fluctuations affecting local businesses.
- Need for accurate and updated historical information.

Figure 2: Visitors exploring a heritage site through a virtual tour.

3. Analyze the Current Approach

Before proposing a new software solution, it is important to understand how heritage tourism services are currently managed. Most heritage sites still use traditional methods such as brochures, guidebooks, information boards, tour guides, and basic tourism websites to provide information to visitors. While these methods are useful, they do not meet the expectations of modern tourists who prefer digital and interactive services. The lack of an integrated platform makes travel planning difficult and reduces visitor convenience. Analyzing the existing system helps identify its strengths, limitations, and the improvements required to develop a better software solution.

3.1 Describe the Existing System or Current Process

The current heritage tourism system mainly depends on physical visits and traditional sources of information. Tourists collect details from travel agencies, guidebooks, tourism websites, or social media before visiting a site. At the heritage location, they rely on signboards, brochures, museum displays, and tour guides for historical information. Ticket booking is usually done at the entrance or through separate booking websites. Most heritage websites provide only basic details and lack advanced features such as virtual tours, AI assistance, multilingual support, and personalized recommendations. Administrative tasks are also handled manually, making the overall process less efficient.

Key Points:

- Information is provided through brochures, signboards, and tour guides.
- Tourists use travel agencies, websites, and social media for planning.
- Ticket booking is mostly manual or through separate websites.
- Heritage websites offer only basic information.
- No integrated platform for all tourism services.
- Administrative work is largely manual.

3.2 Identify the Strengths of the Existing System

The existing system has several strengths that continue to support tourism and heritage preservation. Physical visits allow tourists to experience the beauty and cultural significance of heritage sites directly. Tour guides provide detailed explanations and answer visitors' questions, making the experience more informative. Printed brochures and information boards are useful for visitors without internet access, while official tourism websites help tourists plan their visits by providing essential information such as ticket prices and opening hours. The current system also supports local businesses and creates employment opportunities in the tourism sector.

Key Points:

- Provides real-life heritage experience.
- Tour guides offer detailed historical explanations.
- Brochures and signboards work without internet.
- Official websites provide basic travel information.
- Supports local employment and tourism economy.
- Helps preserve cultural heritage.

3.3 Identify the Limitations of the Existing System

Although effective in some areas, the current system has several limitations. Printed materials cannot be updated easily, and many websites provide only limited information. Tourists often face language barriers, long ticket queues, and a lack of personalized guidance. There are no virtual tour facilities for people who cannot travel, and visitors must use multiple platforms for booking, navigation, and information. Manual administrative work also increases workload and reduces efficiency. These limitations affect the overall visitor experience and highlight the need for digital transformation.

Key Points:

- Limited and outdated information.
- Language barriers for international tourists.
- Long waiting time for ticket booking.
- No virtual tour facilities.
- Manual administration increases workload.
- Multiple platforms make planning difficult.

3.4 Analyze the Problems or Gaps That Need to Be Addressed

The current tourism system lacks a centralized platform that combines all essential services in one place. Tourists need better access to virtual tours, online booking, AI-based assistance, multilingual support, and real-time information. Heritage authorities require efficient tools to manage digital content and visitor records. The system should also improve security, accessibility, and data management while supporting future technologies such as Artificial Intelligence and Virtual Reality. Addressing these gaps through an integrated software solution will enhance visitor satisfaction, improve operational efficiency, and promote sustainable heritage tourism.

Key Points:

- No centralized tourism platform.
- Need for virtual tours and AI assistance.
- Better content and visitor management required.
- Improve security and accessibility.
- Support future technologies like AI and VR.
- Enhance overall tourism experience.

Figure 3: Major stakeholders involved in the Smart Heritage Platform.

4. Propose a Suitable Solution Using Software Engineering Principles

The limitations of the current system show the need for a smart and integrated software solution. The proposed **Smart Heritage Site Information and Virtual Tour Platform** is a web and mobile application that provides heritage information, virtual tours, online ticket booking, AI chatbot support, and GPS navigation in one place. It helps tourists access services easily while enabling administrators to manage content and visitor records efficiently. The platform uses modern technologies such as Artificial Intelligence (AI), Virtual Reality (VR), cloud computing, and secure databases to improve user experience and support heritage conservation.

4.1 Design a Software-Based Solution for the Given Problem

The proposed system provides a centralized platform where users can register, search for heritage sites, view historical information, experience virtual tours, and book tickets online. Multimedia content such as images, videos, and audio guides makes learning more interactive. An AI chatbot assists users by answering queries and providing travel recommendations. Administrators can update heritage information, manage users, monitor bookings, and generate reports through a secure dashboard. This integrated solution improves efficiency, reduces manual work, and enhances visitor satisfaction.

Key Points:

- Centralized web and mobile platform.
- User registration and secure login.
- Heritage information with multimedia content.
- Virtual tour facility.
- Online ticket booking and payment.
- AI chatbot for user support.
- Admin dashboard for system management.

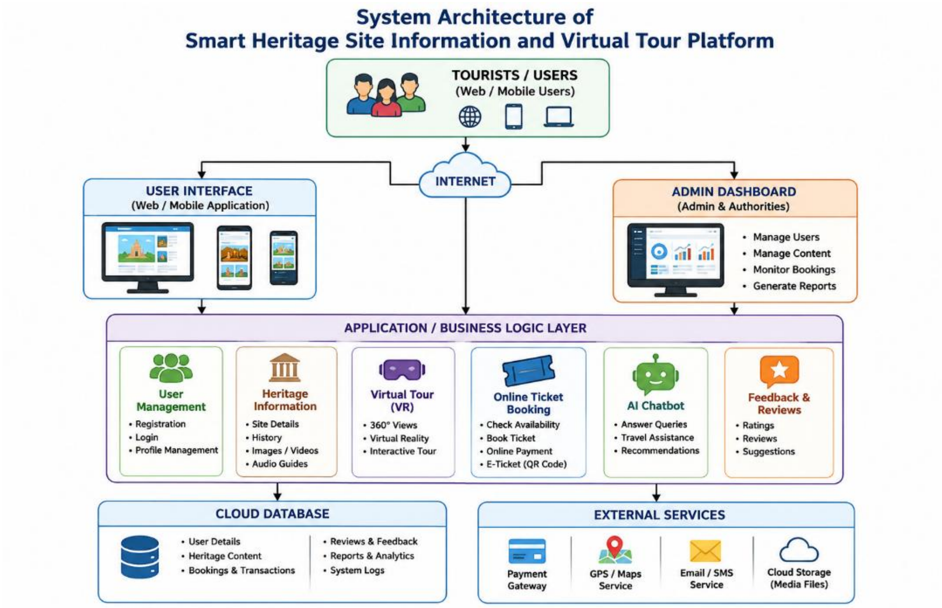

Figure 4.1: System Architecture of the Smart Heritage Site Information and Virtual Tour Platform

4.2 Describe the Major Functional Modules of the Proposed System

The platform consists of several modules that work together to provide complete tourism services. The User Management module handles registration and login. The Heritage Information module stores details about historical sites. The Virtual Tour module offers immersive 360-degree experiences. The Ticket Booking module manages reservations and payments. The AI Chatbot provides instant assistance, while the Feedback module collects user opinions. The Administration module manages users, content, and reports to ensure smooth system operation.

Key Points:

- User Management Module.
- Heritage Information Module.
- Virtual Tour Module.
- Online Ticket Booking Module.
- AI Chatbot Module.
- Feedback and Review Module.
- Administration Module.

4.3 Identify the Functional and Non-Functional Requirements

The functional requirements define what the system should do, such as user registration, login, heritage search, virtual tours, online booking, AI assistance, and feedback submission. Administrators should be able to manage users, update content, and monitor the platform. Non-functional requirements ensure software quality by providing good performance, security, reliability, scalability, maintainability, and ease of use. These requirements help the platform deliver a secure, efficient, and user-friendly experience.

Key Points:

Functional Requirements

- User registration and login.
- Heritage site search.
- Virtual tours.
- Online ticket booking.
- AI chatbot assistance.
- Feedback and review system.
- Admin management.

Non-Functional Requirements

- High performance.
- Reliable system operation.
- Secure authentication and data protection.
- Scalability.
- Easy maintenance.
- User-friendly interface.

4.4 Explain How Software Engineering Principles Are Incorporated into the Solution

The proposed platform follows important software engineering principles to ensure high quality and long-term success. Modularity divides the system into independent modules, making development and maintenance easier. Scalability allows the platform to support more users and additional heritage sites. Maintainability ensures that updates and bug fixes can be implemented efficiently. Reliability provides stable system performance with minimal downtime. Usability offers a simple and responsive interface for all users, while security protects user data through encrypted passwords, secure payment systems, and role-based access control. These principles make the platform efficient, flexible, and future-ready.

Key Points:

- **Modularity** – Independent system modules.
- **Scalability** – Supports future growth.
- **Maintainability** – Easy updates and debugging.
- **Reliability** – Stable and continuous performance.
- **Usability** – Simple and responsive interface.
- **Security** – Data encryption and secure access.

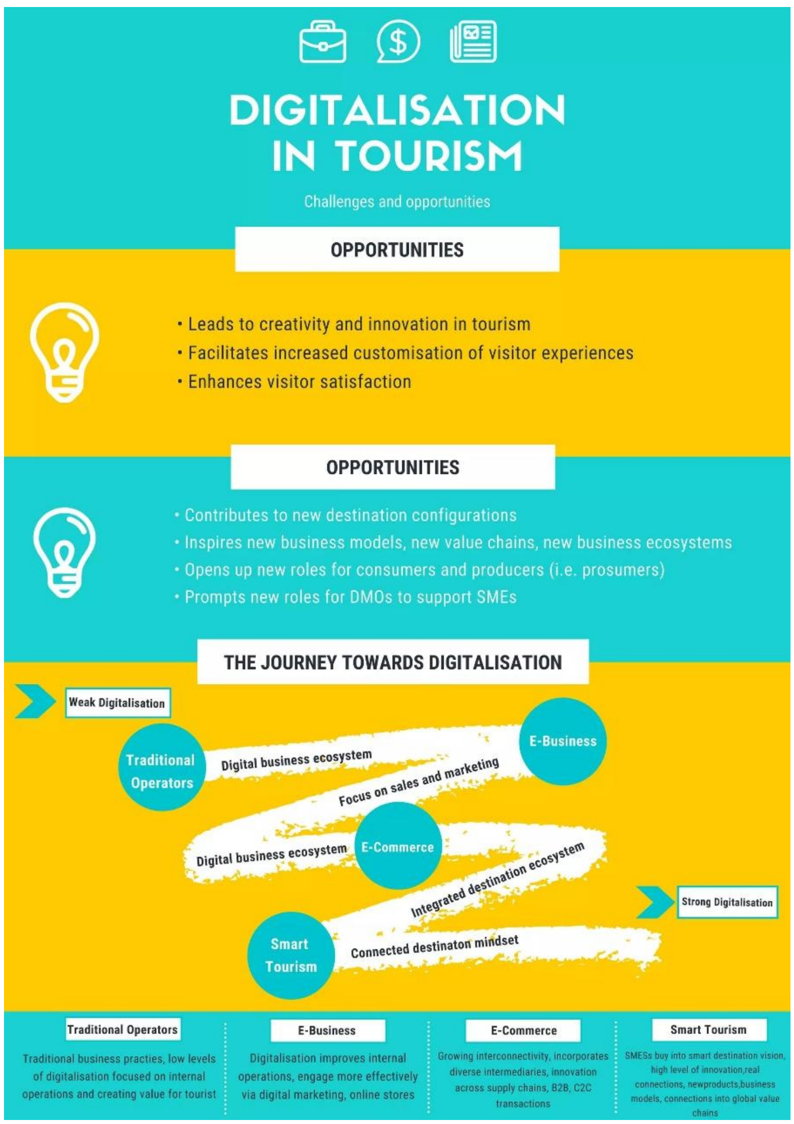

Figure 4: Comparison between the existing tourism system and the proposed smart tourism platform.

5. Justify the Chosen Methodology/Framework

Selecting the right Software Development Life Cycle (SDLC) model is essential for building a reliable and efficient software system. The Smart Heritage Site Information and Virtual Tour Platform contains multiple modules, such as user management, virtual tours, ticket booking, AI chatbot, and content management. Since user requirements may change over time, the project requires a flexible and adaptive development approach. **Agile Scrum** is selected because it supports iterative development, continuous testing, and regular stakeholder feedback. This methodology helps deliver high-quality software while allowing improvements throughout the development process.

5.1 Select an Appropriate SDLC Model

The **Agile Scrum** model is chosen because it divides the project into small development cycles called **sprints**. Each sprint focuses on developing and testing a specific module of the system. At the end of every sprint, the completed work is reviewed, and feedback is collected before starting the next phase. This approach helps identify errors early, improves software quality, and allows the system to adapt to changing requirements. Agile Scrum is well suited for projects that require continuous updates and feature enhancements.

Key Points:

- Selected SDLC Model: **Agile Scrum**
- Development is divided into small sprints.
- Each sprint develops specific features.
- Regular review and stakeholder feedback.
- Early error detection and testing.
- Supports continuous improvement.

5.2.Justify Why the Selected Methodology is Suitable

Agile Scrum is suitable for this project because heritage information and tourism services require regular updates. The platform may need new features, improved content, and technology upgrades over time. Agile allows developers to implement these changes without affecting the entire system. Continuous communication between developers, administrators, and stakeholders ensures that user requirements are met effectively. Regular testing during each sprint also improves software quality and minimizes development risks.

Key Points:

- Flexible and adaptable development.
- Supports changing user requirements.
- Continuous stakeholder involvement.
- Faster delivery of software features.
- Regular testing improves quality.
- Reduces project risks.

5.3 Explain How It Supports Successful Project Development and Maintenance

Agile Scrum supports successful project development through teamwork, regular communication, and continuous testing. Each sprint produces a working version of the software that can be reviewed and improved based on stakeholder feedback. After deployment, the system can be easily maintained by releasing updates, fixing bugs, and adding new features without affecting existing modules. This approach ensures that the platform remains reliable, secure, and scalable while meeting the evolving needs of users and the tourism industry.

Key Points:

- Encourages teamwork and collaboration.
- Delivers working software in every sprint.
- Easy maintenance and software updates.
- Quick bug fixing and feature enhancements.
- Improves reliability and scalability.
- Ensures long-term system success.
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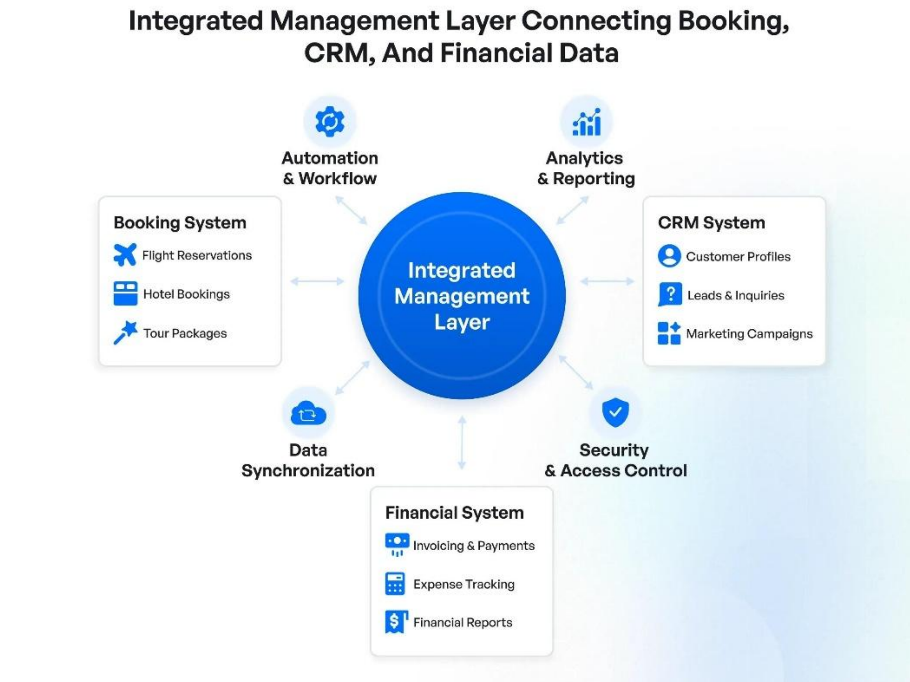

Figure 5: Architecture of the Smart Heritage Site Information and Virtual Tour Platform.

6. Discuss Expected Outcomes and Limitations

The proposed **Smart Heritage Site Information and Virtual Tour Platform** is expected to improve the tourism experience by providing a modern and user-friendly digital solution. It offers features such as virtual tours, online ticket booking, AI chatbot support, and multilingual services through a single platform. The system also helps heritage authorities manage information efficiently and preserve cultural heritage digitally. By combining advanced technologies with software engineering principles, the platform promotes sustainable tourism and improves accessibility for users worldwide.

6.1 Describe the Expected Benefits of the Proposed Solution

The proposed platform provides several benefits to tourists, heritage authorities, and the tourism industry. Tourists can easily access accurate information, explore virtual tours, and book tickets online without waiting in long queues. Heritage authorities can efficiently manage digital content, visitor records, and reports. The platform also supports students and researchers by providing reliable historical information. In addition, it reduces paperwork, improves operational efficiency, and promotes cultural awareness through digital technologies.

Key Points:

- Easy access to heritage information.
- Virtual tours for remote visitors.
- Secure online ticket booking.
- Better content and visitor management.
- Supports education and research.
- Promotes sustainable tourism.

6.2 Explain How the Solution Addresses the Identified Challenges

The proposed solution overcomes the limitations of the existing system by integrating all tourism services into one platform. It provides heritage information, ticket booking, virtual tours, AI assistance, and multilingual support in a single application. Cloud technology enables real-time updates, while secure authentication protects user information. The administrator dashboard simplifies content management and reduces manual work. Overall, the platform improves accessibility, efficiency, and user satisfaction.

Key Points:

- Centralized tourism platform.
- Virtual tours improve accessibility.
- AI chatbot provides instant support.
- Multilingual support for global users.
- Real-time information updates.
- Secure and efficient system management.

6.3 Discuss Potential Risks, Implementation Challenges, and Limitations

Although the platform offers many advantages, some challenges may arise during implementation. Developing advanced technologies such as AI and Virtual Reality requires high investment and technical expertise. The system also depends on a stable internet connection, which may not be available in remote areas. Regular maintenance, cybersecurity, and updating historical information are essential for smooth operation. In addition, virtual tours cannot completely replace the real experience of visiting heritage sites.

Key Points:

- High development and implementation cost.
- Dependence on internet connectivity.
- Cybersecurity and data privacy concerns.
- Continuous content updates required.
- Regular system maintenance needed.
- Virtual tours cannot fully replace physical visits.

6.4 Suggest Possible Future Enhancements

The platform can be improved further by integrating advanced technologies in future versions. Augmented Reality (AR) can provide interactive information during physical visits, while advanced AI can offer personalized travel recommendations. Blockchain technology can improve ticket security, and IoT devices can support smart visitor management. Additional features such as voice assistants, live virtual events, and predictive analytics can further enhance the user experience and make the platform more efficient.

Key Points:

- Integrate Augmented Reality (AR).
- Advanced AI-based recommendations.
- Blockchain for secure ticketing.
- IoT for smart visitor management.
- Voice assistant support.
- Predictive analytics and live virtual events.

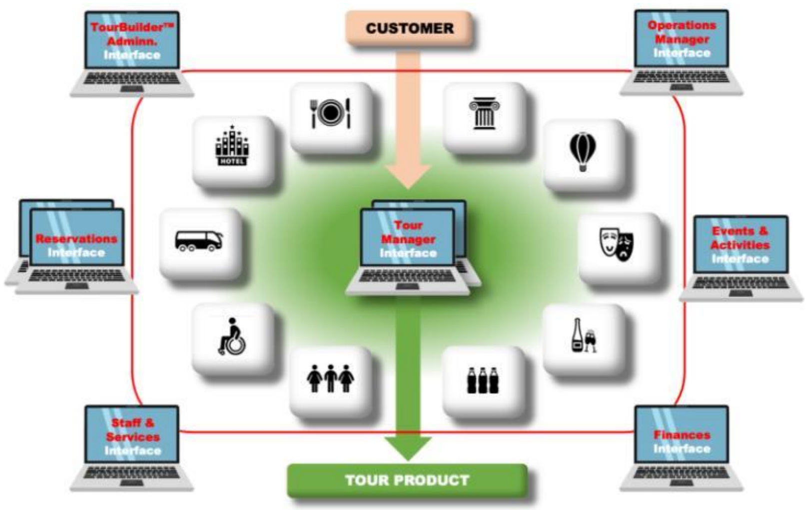

Figure 6: Functional modules of the proposed system.

Conclusion

The Smart Heritage Site Information and Virtual Tour Platform provides a modern and effective solution for improving heritage tourism through digital technology. By integrating features such as virtual tours, online ticket booking, AI-based assistance, and multilingual support, the platform overcomes the limitations of traditional tourism systems and enhances the overall visitor experience. The adoption of the Agile Scrum methodology ensures flexible development, continuous improvement, and easy maintenance, while software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security make the system efficient and future-ready. Overall, the proposed solution supports cultural heritage preservation, promotes sustainable tourism, and contributes to the growth of the tourism and hospitality industry by making heritage information more accessible and engaging for users around the world.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5